

Hospital Management System

Course Title : Introduction to Programming Language II LAB

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Group Members

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Hospital Management System

Introduction

The Hospital Management System (HMS) is designed for Any Hospital to replace their existing manual, paper based system. This System targets to provide complete solution for Hospital and Health care services to manage Admin panel . This System can be used in any Hospital, Clinic, Diagnostics or Pathology labs for maintaining patient details, doctor details, nurse details and their test results & reports . It integrates the entire Resources of a Hospital Manegment into One Integrated Software Application.

Objectives

The objective of this project is to let the students apply the programming knowledge into a real world situation/problem and exposed the students how programming skills helps in developing a good software.

>>To develop a Java-based Hospital Management System that provides an organized solution for managing essential hospital records.

>>To apply Object-Oriented Programming principles and Java programming techniques in the development of a practical real-world software project.

>>To implement key functionalities including patient, doctor, nurse, and billing management, along with CRUD operations, file handling, exception handling, and report generation.

Motivation

The main motivation for selecting the Hospital Management System as our project topic was to apply programming concepts to a practical real-world problem. Hospitals need to manage different types of information, such as patient, doctor, nurse, and billing records. They managed it manually, It's Maintaining and searching records manually requires more time, Paper-based records can be lost, damaged, or misplaced and Records are not easily accessible or updated from a centralized system. Calculating total bills, paid amounts, and due amounts manually may cause calculation errors. So, these are the reason why we Developing this system allowed us to practice Object-Oriented Programming, Java GUI development, CRUD operations, file handling, and exception handling while creating a useful software solution.

Applications

The Hospital Management System can be used in small hospitals, clinics, and healthcare centers to manage basic hospital records and daily administrative activities.

The Program will use Hospital administrators Receptionists and front-desk staff Doctors Nurses Billing or administrative staff.

This program benefits :

- > Store, update, view, and delete patient information.
- > Maintain doctor specialization and nurse shift information.
- > Record patient bills, paid amounts, and calculate due amounts.
- > Save hospital records in files for future use.
- > Generate a report containing important hospital information.
- > Makes record management faster and more organized.

Project Description

SOFTWARE DEVELOPMENT PROCESS

The project requirements were analyzed and organized into the following functional modules:

1. Patient Management Module

- Patient record creation and registration
- Patient record viewing and updating
- Patient record deletion
- Patient billing information management

2. Doctor Management Module

- Doctor record creation and registration
- Doctor information viewing and updating
- Doctor record deletion
- Doctor specialization management

3. Nurse Management Module

- Nurse record creation and registration
- Nurse information viewing and updating
- Nurse record deletion
- Nurse shift management

4. Billing and Report Module

- Patient identification through patient ID
- Total bill and paid amount management
- Automatic calculation of outstanding due amount
- Generation of hospital reports

5. Data Management Module

- Storing records using file handling
- Loading previously stored records
- Maintaining separate files for different types of records

6. Graphical User Interface Module

- User login and authentication
- Navigation through different system modules
- Interactive data entry and management using Java Swing

DEVELOPMENT METHODOLOGY

Phase 1

- Model Classes

Phase 2

- CRUD & File Handling

Phase 3

- Exception Handling

Phase 4

- Billing & Report Management

Phase 5

- Graphical User Interface Integration (GUI)

Phase 6

- Testing and Bug Fixing

SYSTEM DESIGN

Package Structure

src

|— Person

|— Patient

|— Doctor

|— Nurse

|— InvalidDataException

|— SimpleHospitalApp

|— Main

Person —

- Abstract base class for common person information.

Patient —

- Manages patient information and billing details.

Doctor —

- Manages doctor information and specialization.

Nurse —

- Manages nurse information and shift details.

InvalidDataException —

- Handles invalid data using a custom exception.

SimpleHospitalApp —

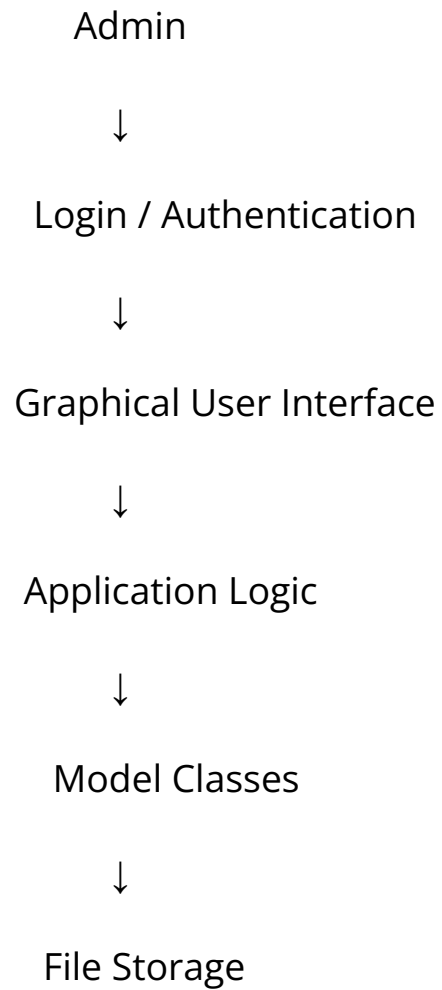
- Contains the GUI, system operations, file handling, and report generation.

Main —

- Application entry point.

SYSTEM ARCHITECTURE

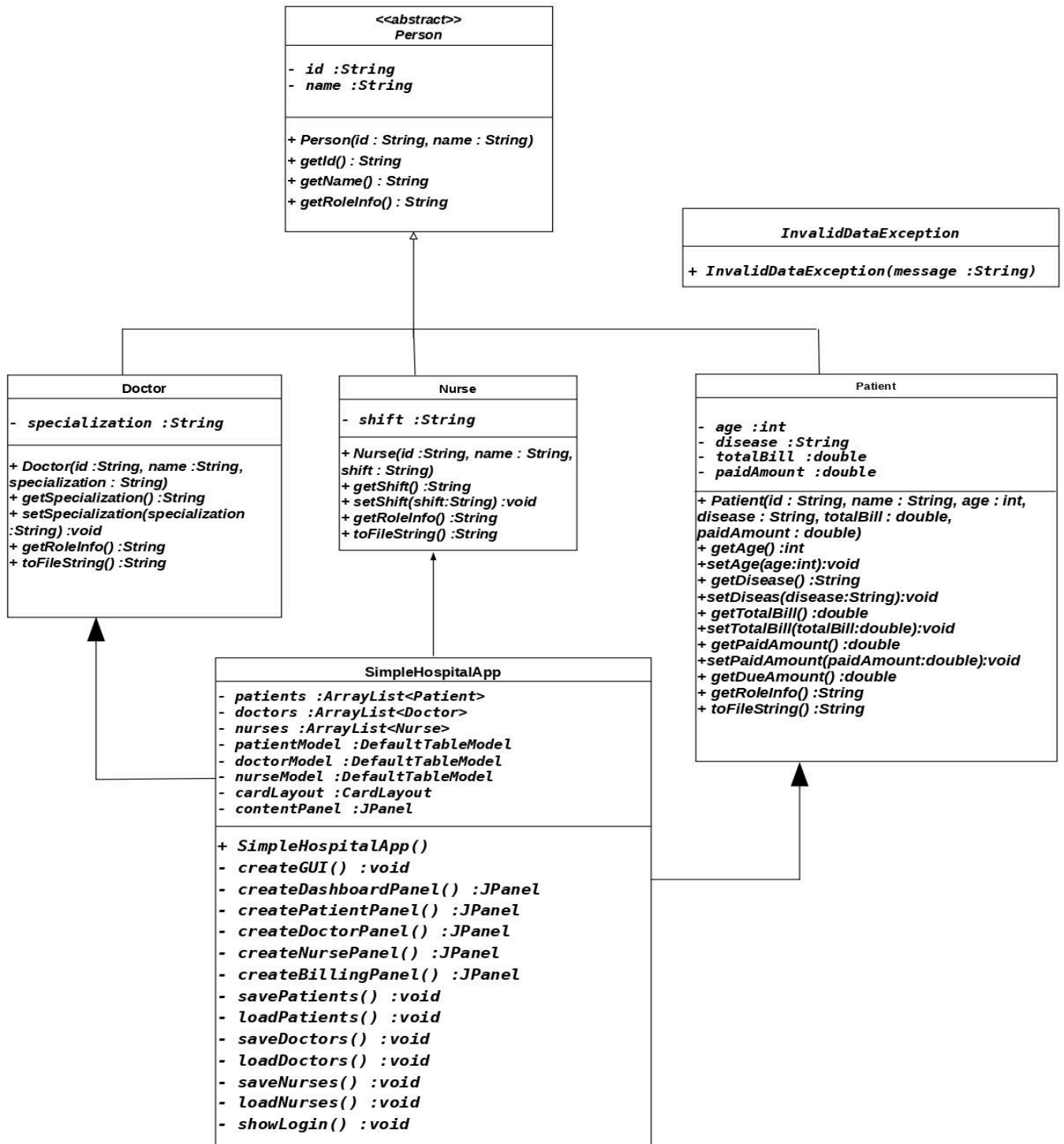
The Hospital Management System follows a simple layered architecture:



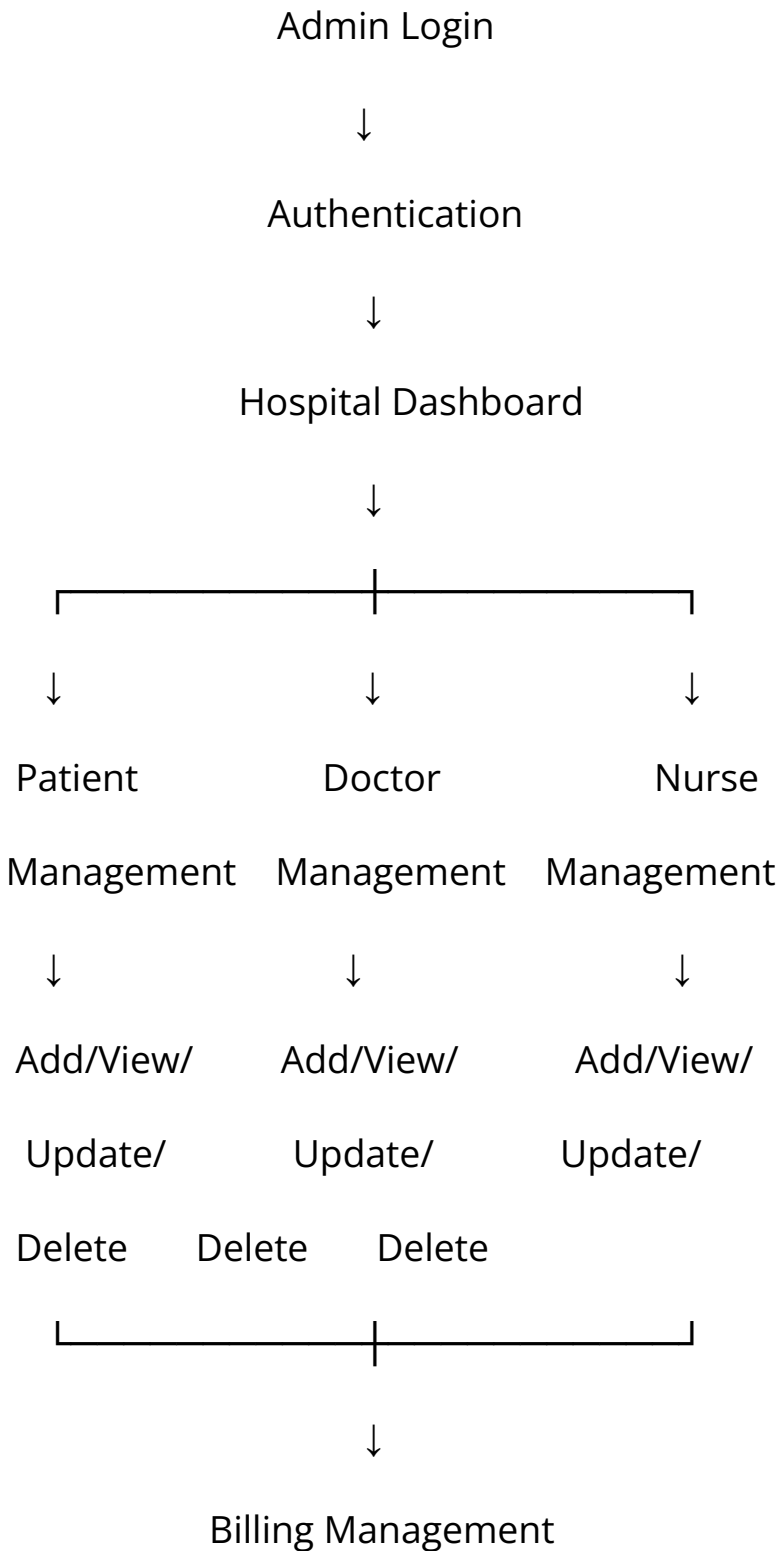
Class Design

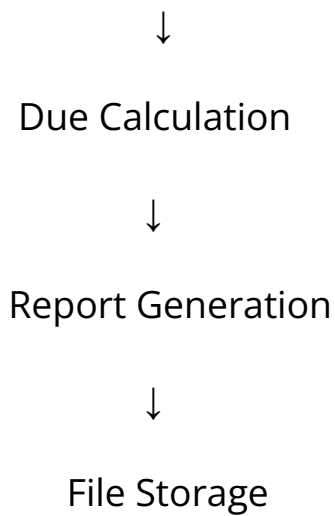
CLASS	RESPONSIBILITY
Person	Abstract base class containing common information and behavior for patients, doctors, and nurses.
Patient	Manages patient information, disease details, and billing information.
Doctor	Manages doctor information and specialization details.
Nurse	Manages nurse information and shift details.
SimpleHospitalApp	Controls the GUI, CRUD operations, billing, file handling, and report generation.
InvalidDataException	Handles invalid data through a custom exception.
Main	Serves as the entry point of the application.

Project Diagram



Work Flow





OOP Concepts Use :

➤ Encapsulation:

Encapsulation was achieved by declaring class attributes as private and providing controlled access through public methods.

Classes demonstrating encapsulation:

- Person
 - Patient
 - Doctor
 - Nurse
-

➤ Inheritance:

Inheritance was implemented by extending the common **Person** class.

Person

|

├── Patient

├── Doctor

└── Nurse

Parent Class:

- **Person**

Child Classes:

- **Patient**
- **Doctor**
- **Nurse**

➤ Abstraction:

The **Person** class was designed as an abstract class that defines common properties and behavior for different types of hospital personnel and patients.

>>Abstract Class:

- `Person`

>>Abstract Method:

- `getRoleInfo()`

>>Concrete Implementations:

- `Patient.getRoleInfo()`
 - `Doctor.getRoleInfo()`
 - `Nurse.getRoleInfo()`
-

➤ **Polymorphism:**

Polymorphism was implemented through method overriding.

Examples:

- `Patient.getRoleInfo()`
- `Doctor.getRoleInfo()`
- `Nurse.getRoleInfo()`

Objects of different subclasses can be treated as `Person` objects while executing their own specialized implementations of `getRoleInfo()`.

➤ Exception Handling:

Custom exception handling was implemented to validate data and handle invalid input during system operations.

Custom Exception:

- `InvalidDataException`

Exception Handling:

- `try-catch` blocks are used to handle invalid input and file-related errors.

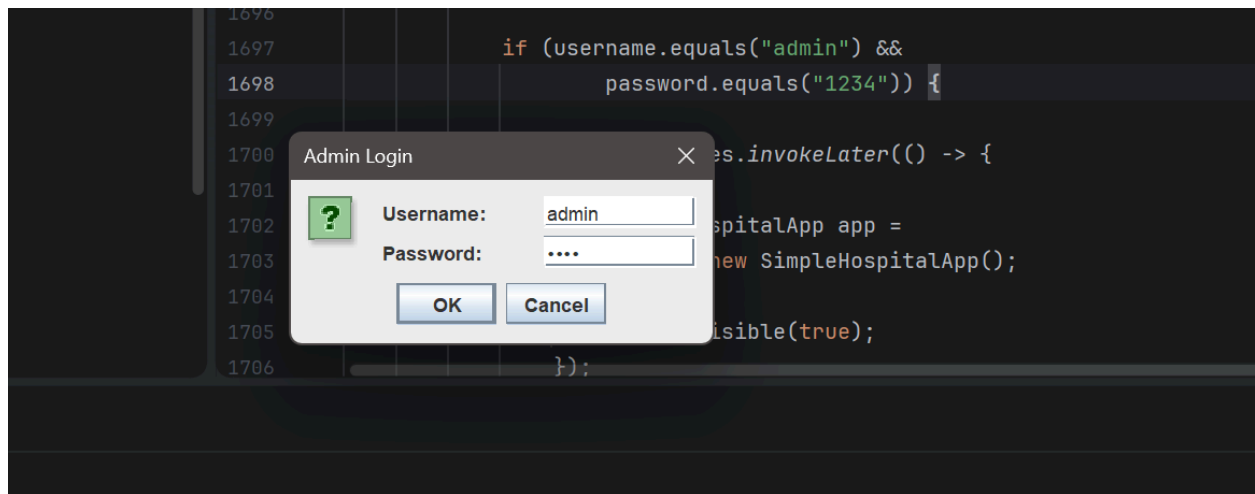
TOOLS & TECHNOLOGIES

- Programming Language: Java
- Java Version: Java 17 or later
- Programming Paradigm: Object-Oriented Programming (OOP)
- IDE: IntelliJ IDEA
- Build/Execution: Java compiler and IntelliJ IDEA's built-in execution environment
- Version Control: Git
- Repository: GitHub

Libraries / Frameworks Used:

- Java Swing — for developing the graphical user interface.
- Java AWT — for GUI components and layout management.
- Java Collections Framework — **ArrayList** for managing records.
- Java I/O — for file handling and data persistence.

GUI SCREENSHOTS /OUTPUT



Hospital Management System

HOSPITAL MANAGEMENT SYSTEMAdmin Panel

Dashboard

Patients

Doctors

Nurses

Billing & Report

Exit

Patient Management

ID	Name	Age	Disease	Total Bill	Paid	Due
2024200000100	Rakib	45	Fever	10000.0	8000.0	2000.0
2024200000101	Mehedi	22	Dengue	5000.0	5000.0	0.0
2024200000102	Shanto	35	Diarrhea	15000.0	12000.0	3000.0
2024200000103	Nobi	30	Typhoid	30000.0	29000.0	1000.0
2024200000104	Anik	25	Pneumonia	1000.0	200.0	800.0

Add Patient

Update Patient

Delete Patient

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Doctor Management

ID	Name	Specialization
2024200000200	Rahim	Cardiologist
2024200000201	Sabbir	Neurologist
2024200000202	Karim	Dermatologist
2024200000203	Karib	Gastroenterologist
2024200000204	Farid	Orthopedic

Add Doctor

Update Doctor

Delete Doctor

Hospital Management System

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Billing & Report

Exit

Nurse Management

ID	Name	Shift
2024200000300	Akhl	Day
2024200000301	Kona	Day
2024200000302	Mim	Night
2024200000303	Nusrat	Day
2024200000304	Anika	Night

Add Nurse

Update Nurse

Delete Nurse

Hospital Management System

HOSPITAL MANAGEMENT SYSTEMAdmin Panel

Dashboard

Patients

Doctors

Nurses

Billing & Report

Exit

Billing & Report

Patient ID: 2024200000100

Search Bill

Generate Report

Patient ID : 2024200000100

Patient Name : Rakib

Disease : Fever

Total Bill : 10000.0

Paid Amount : 8000.0

Due Amount : 2000.0



Contribution :

Each group member contributed to the development of the Hospital Management System by taking responsibility for specific modules and project components.

SL	Group Member	ID	Tasks Completed	Modules / Features Developed
1	MD RAKIBUL ISLAM (Leader)	2024200000352	Coordinated the project development and implemented billing-related operations and report generation.	Billing & Report Module • Patient billing management • Due amount calculation • Hospital report generation • CRUD Operations
2	Mehedi Hasan	2024200000187	Developed and implemented the doctor-related class and its management functionality.	Doctor Management Module • Doctor class • Doctor information management • Specialization management • CRUD operations
3	Mohammad Hasibul Islam Shanto	2025000000210	Developed and implemented the nurse-related class and its management functionality.	Nurse Management Module • Nurse class • Nurse information management • Shift management • CRUD operations
4	Md. Oly Islam Nobil	2023200000603	Developed and implemented the patient-related class and its management functionality.	Patient Management Module • Patient class • Patient information management • Disease and billing information • CRUD operations
5	Syed Nazim Mahmud Anik	2023200000280	Prepared and organized the major system diagrams and supported the project documentation.	System Design & Documentation • Class Diagram • System Architecture • System Workflow • Project documentation

Conculation

The Hospital Management System was successfully designed and implemented using Java and Object-Oriented Programming principles. The system provides basic functionalities for managing patient, doctor, and nurse records, along with billing, file handling, exception handling, and report generation.

The project successfully achieved its primary objectives by applying Encapsulation, Inheritance, Abstraction, and Polymorphism in a practical software project. It also provided practical experience in Java Swing GUI development, CRUD operations, file handling, testing, and collaborative development using Git and GitHub.

The system has some limitations, such as using text files instead of a database and providing basic authentication. Future improvements may include database integration, role-based access control, appointment management, medicine management, advanced search, and improved report generation.

Overall, the project provided practical experience in software design, implementation, testing, Object-Oriented Programming, and team-based software development.

Github Project Link :

<https://github.com/DarkRakib11/Hospital-Manegment-System>